

■■ INDIAN SPACE RESEARCH ORGANISATION (ISRO) — SAC AHMEDABAD

COMPONENT-WISE SPACEFLIGHT QUALIFICATION CERTIFICATE
Compliance: MIL-STD-883 Method 1015 | MIL-PRF-38535 Class V | ISRO PS #26170

1. Component Metadata & Environmental Stress Profile

Component Serial ID:	ISRO-SAC-2026-0001	Test Chamber Temp:	125.0°C (HTOL Stress)
Device Family:	DIGITAL_IC	Chamber Vacuum:	10■■ Torr (TVAC System)
Project Flight Code:	ISRO_GAGANYAAN_SAC_LOT_01	ATE Station ID:	SAC-AHMEDABAD-ATE-01
Screening Parameter:	Quiescent Current (μA)	Specification USL:	45.0 μA

2. ATE Parametric Telemetry & 95% Conformal Prediction Bounds

Checkpoint	Observed / Forecast Value	Robust Z-Score	95% Conformal Safety Interval
0h Base (Fresh)	11.20 μA	0.00 σ	Pre-Stress Nominal Baseline
24h Chamber Telemetry	12.10 μA	0.45 σ	Live Measured ATE Telemetry
168h Forecast Trajectory	14.80 μA	0.81 σ	[13.50 - 16.10] μA

3. Physics-Informed Degradation Kinetics & SHAP Explainability

Arrhenius Thermal Acceleration Model: Degradation velocity $v_{24} = \frac{I(24h) - I(0h)}{24}$ follows exponential thermal activation $k(T) = A \cdot \exp(-E_a / k_B T)$ with $E_a = 0.68 \text{ eV}$ for silicon gate oxide traps.

Physical Stress Variable	Observed Value	SHAP Attribution	Governing Physical Failure Mechanism
24h Current Measurement	12.10 μA	+0.715 μA	Arrhenius Thermal Leakage Base
Degradation Velocity (v24)	0.0375 μA/hr	+0.469 μA	Time-Dependent Dielectric Breakdown (TDDB)
Population Robust Z-Score	0.45 σ	+0.189 μA	Wafer Periphery Process Variation
Thermal Stress Factor (125°C)	398.15 K	+0.250 μA	Shockley-Read-Hall (SRH) Trap Generation

4. Quality Assurance (QA) Qualification Verdict

■ QUALIFIED FOR 24H FLIGHT RELEASE (AUTO-PASS)
QA VERDICT — FLIGHT PASS (24H EARLY RELEASE): Component ISRO-SAC-2026-0001 exhibits nominal Arrhenius degradation kinetics ($v_{24} = 0.0375 \text{ μA/hr}$). The non-parametric 95% Conformal Upper Bound (16.10 μA) is safely below the Specification USL (45.0 μA) with a 64.2% safety margin. Recommendation: Qualified for immediate 24h early release. Saves 144 chamber hours.

5. ISRO QA Inspector Sign-Off & SHA-256 Audit Seal

Lead QA Reliability Engineer:	_____	Date & Verification Stamp:	_____
ISRO SAC Authority:	_____	SHA-256 Cryptographic Hash:	e3b0c44298fc1c149afb4c8996fb92427ae41e4649b934ca495991b7852b855